

<2018 TEST >

Problem1 [20pt]

- (1) The data of a river has a mean of 600 cms and a standard deviation of 200 cms. Without making any distributional assumption regarding the data, what can be said of the probability that the peak flow in a year selected at random will deviate more than 400 cms from the mean?
- (2) Can you write the Chebyshev inequality in terms of the sample mean of the size N?
- (3) How many observations would be required to be at least 95% sure that the estimated mean peak flow was within 100 cms of its true value if we know nothing of the distribution of peak flows?
- (4) Re-solve the above problem (1) and (3) assuming that the data are normally distribution? Make your comment.

Problem 2 [20pt]

Refer the following theorem: for a Poisson process $N(t)$ of rate λ , the number of event in interval $(a,b]$, is Poisson of mean $\lambda (b-a)$. Assume the number of rainfalls in July in a certain region forms a Poisson process with 1 rainfall every two days in average, find

- (1) Probability [2 rainfall by day three and 5 rainfall by day seven]
- (2) Probability [at most 1 by day three and at least 2 by day seven]
- (3) Probability [the second rainfall occurs before day 7 and third before day 9]

Problem 3 [20pt]

In flood protection planning, the 100-year flood is often used as the design flow.

- (1) In order to be 90% sure that the design flow is not exceeded in a N-year planning period of N, what should be N?
- (2) Calculate the probability of at least one 100-year flood in a 100-year period. Can you also approximate the risk of failure if the design return period is equal to the design life? Make your comment on this finding.

(3) If floods at 2 different sites occur independently, what is the probability of at least one 100-year flood at some site in a 100-year period? Can you re-calculate the probability when two sites are 50% spatially correlated?

Problem 4 [20pt]

Refer to a flow data set of a certain region which has the following observations (unit: cms) from two sites A and B:

- Site A: 7, 9, 2, 16, 9, 19, 5, 10, 3, 8
- Site B: 11, 26, 33, 18, 16, 20

- (1) Assuming that the region is homogeneous in distribution, estimate the 10-year flood for an ungauged site C. What kind of reasonable assumptions do you make for the estimation?
- (2) In the regional flood frequency analysis, explain in Korean a major conceptual difference between the discordance and the heterogeneity tests proposed Hosking and Wallis.
- (3) Explain in Korean the usefulness of the regional skewness map in the US Bulletin 17B.

Good Luck!